

Quantum Engineering Curriculum

Sprint 2 Quiz

Instructions: Answer without running code unless the question explicitly asks you to. Explain your reasoning where appropriate.

1. What is the purpose of the Hadamard (H) gate?
2. Why do repeated executions of an H-gate circuit produce different measurement counts?
3. Explain 'superposition' in your own words.
4. What information does a histogram of counts provide?
5. Write a circuit that places one qubit into superposition and measures it.